

Activation Functions and Non-Linearity

Deep Learning

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Contents of This Video

In this video, we will cover:

- Why linear activations fail for deep networks
- The sigmoid activation function
- The tanh activation function
- The ReLU activation function (most common today)
- Choosing activation functions for different tasks

The Problem with Linear Activations

What if we use a linear activation: $\sigma(z) = z$?

$$\text{Layer 1: } a^{(1)} = W^{(1)}x + b^{(1)}$$

$$\text{Layer 2: } a^{(2)} = W^{(2)}a^{(1)} + b^{(2)}$$

Substituting:

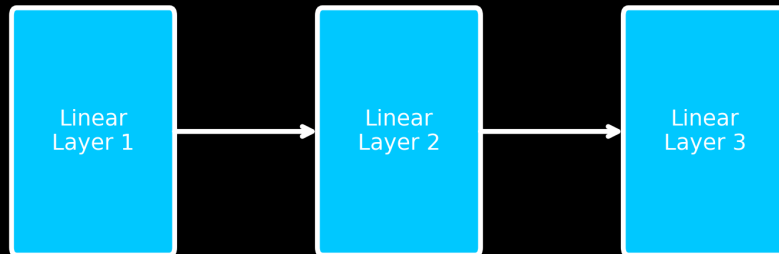
$$a^{(2)} = W^{(2)}(W^{(1)}x + b^{(1)}) + b^{(2)}$$

$$a^{(2)} = \underbrace{(W^{(2)}W^{(1)})}_{W'}x + \underbrace{(W^{(2)}b^{(1)} + b^{(2)})}_{b'}$$

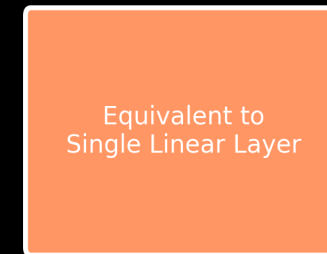
$$a^{(2)} = W'x + b'$$

Linear Compositions Collapse

Multiple Linear Layers



Collapses to One Layer!

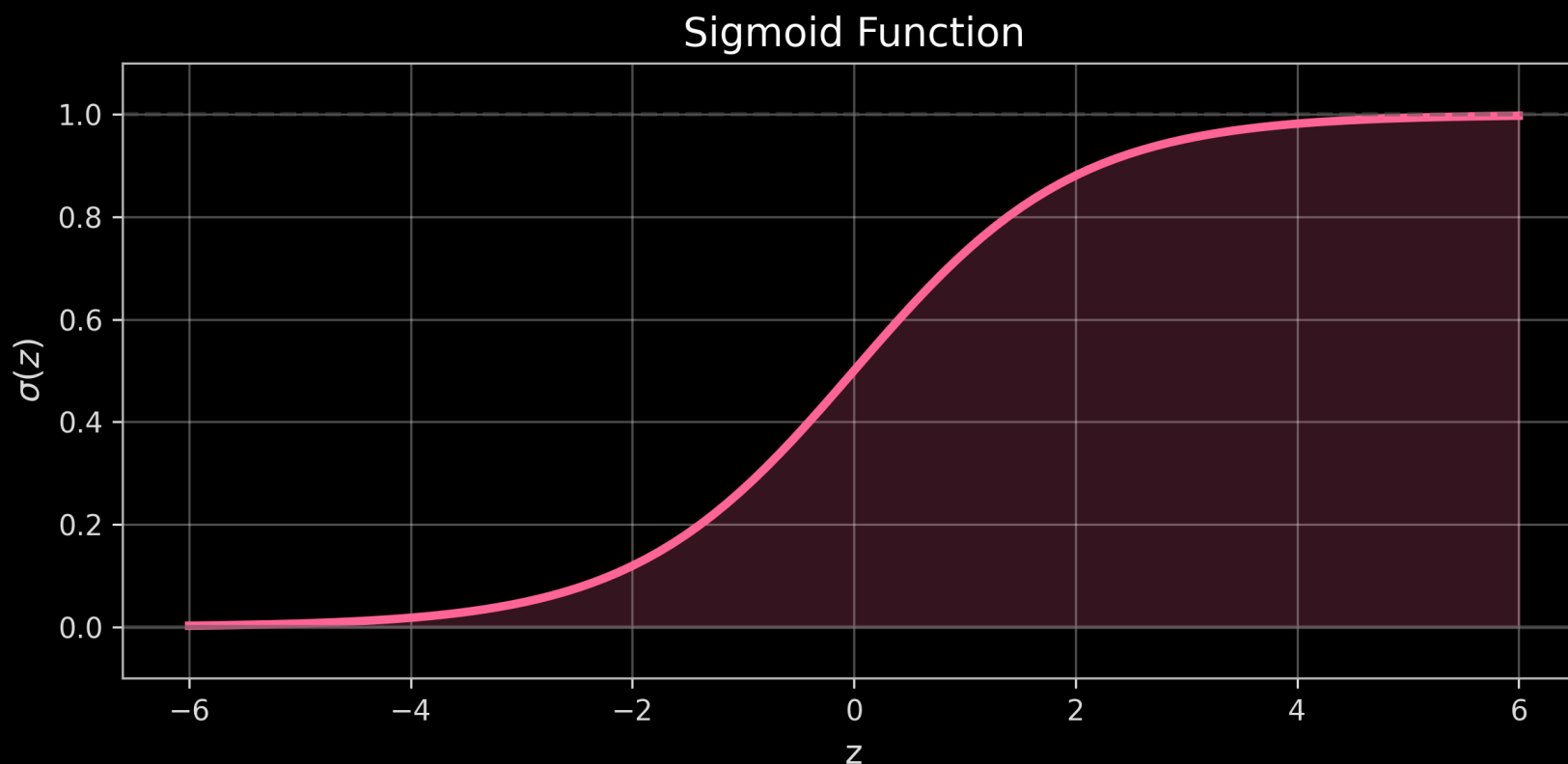


$$W' = W^{(3)}W^{(2)}W^{(1)}$$

Key insight: Without non-linear activations, depth is useless!

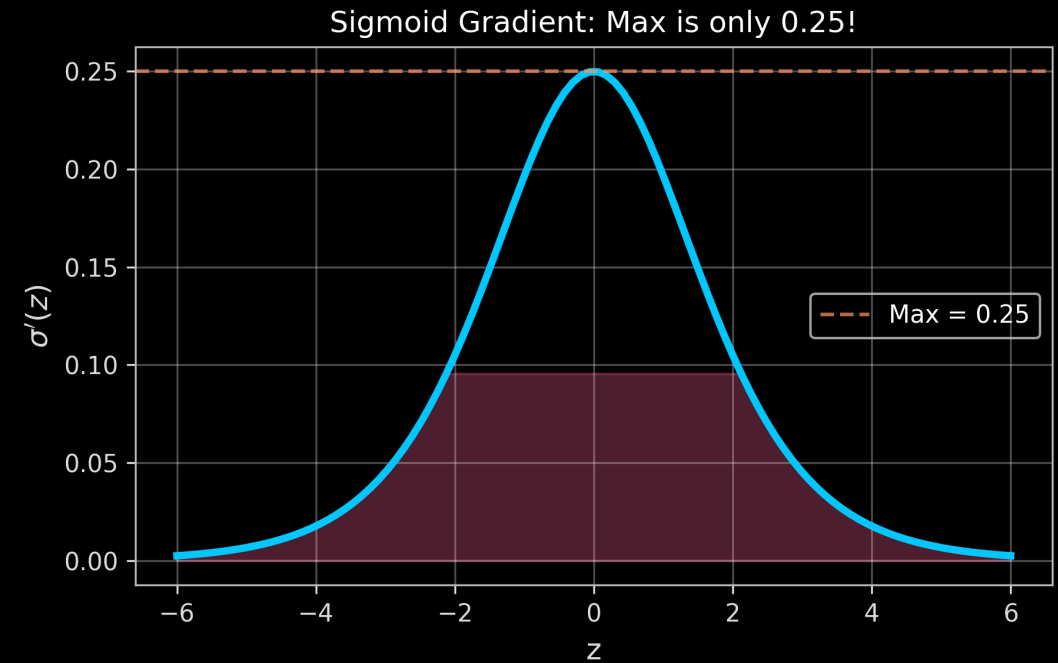
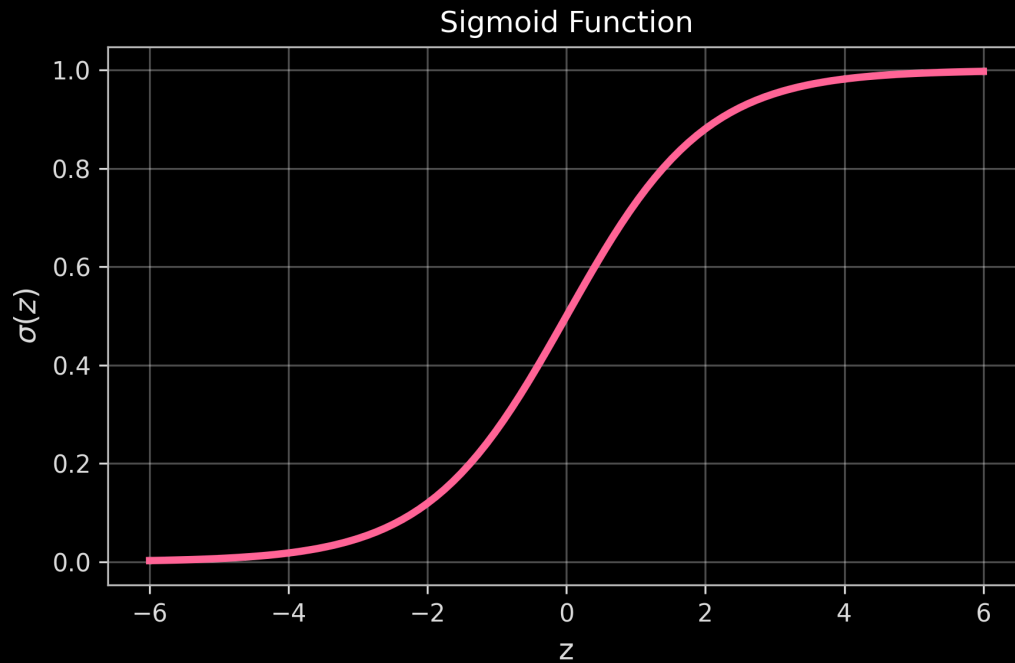
The Sigmoid Activation Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



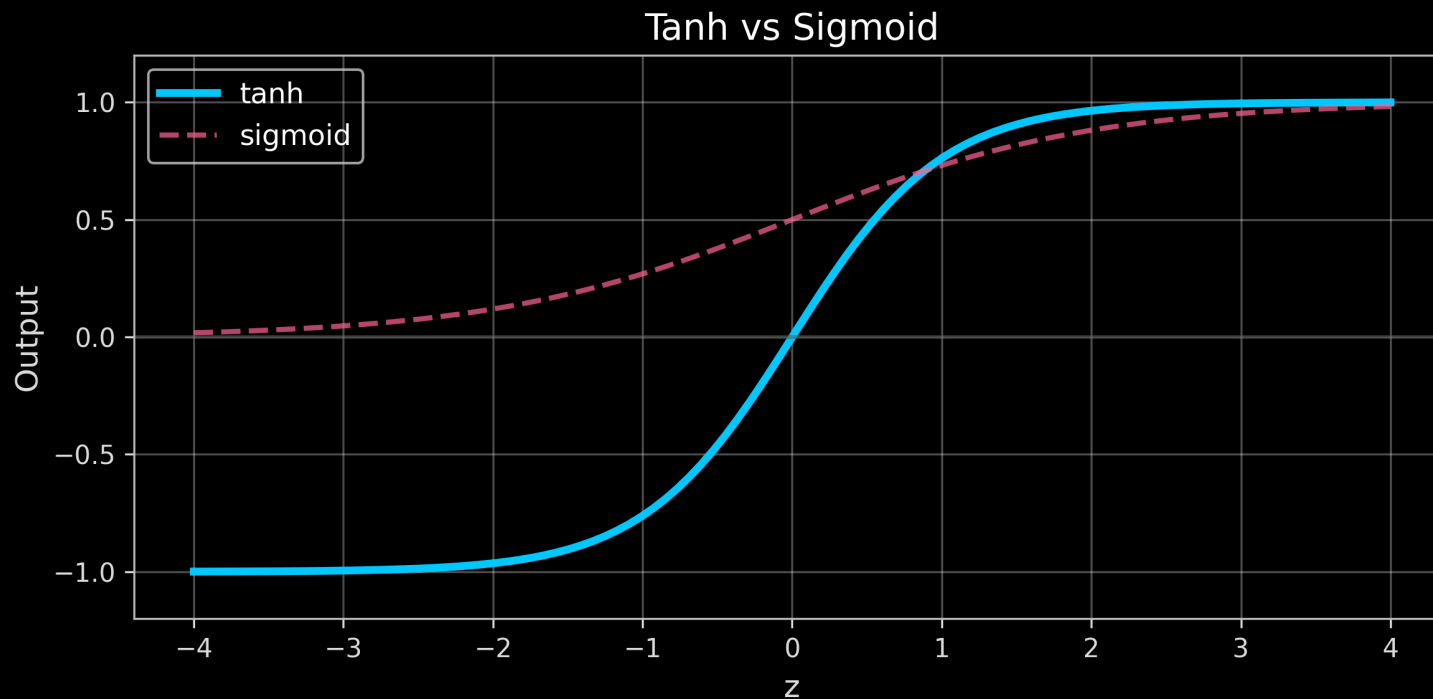
Properties: Output range (0, 1) • Smooth and differentiable • Historically popular

Sigmoid: The Vanishing Gradient Problem



The Tanh Activation Function

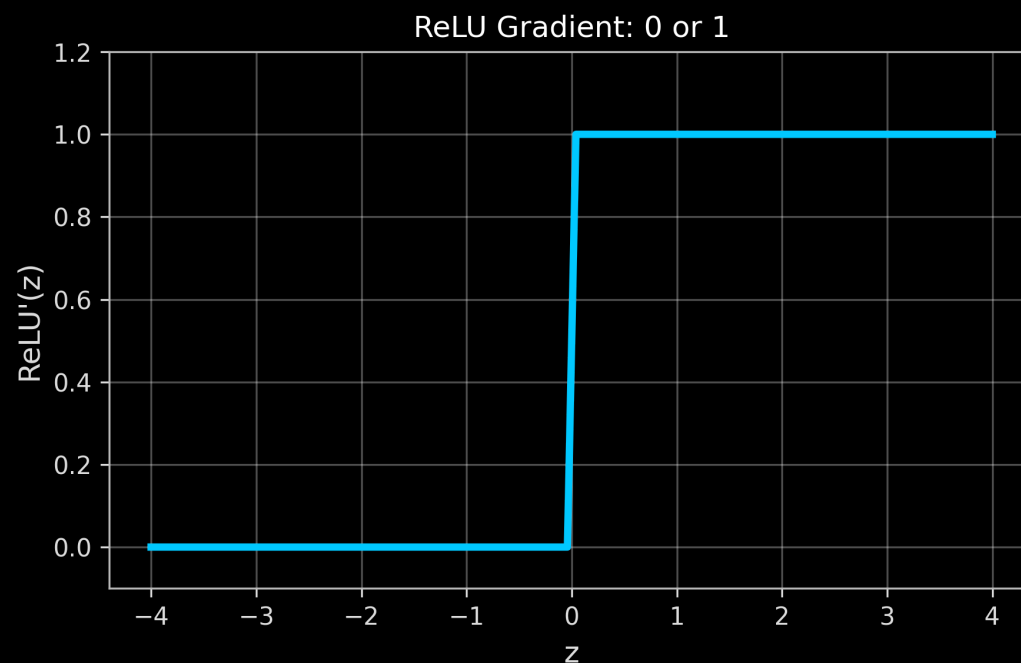
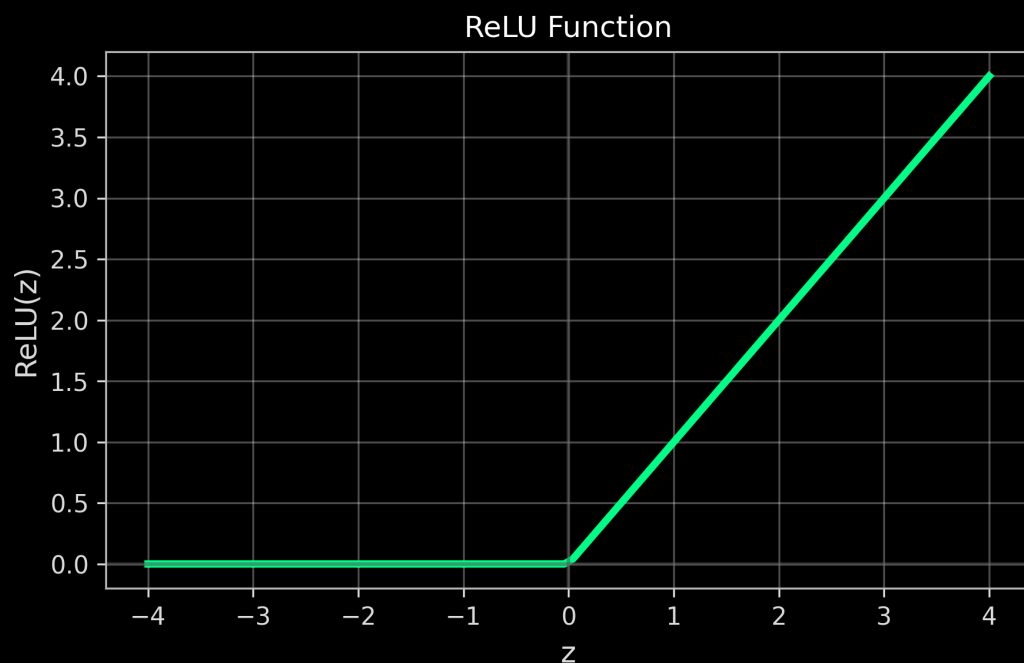
$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$



Properties: Output range (-1, 1) • Zero-centered • Still has vanishing gradient issue

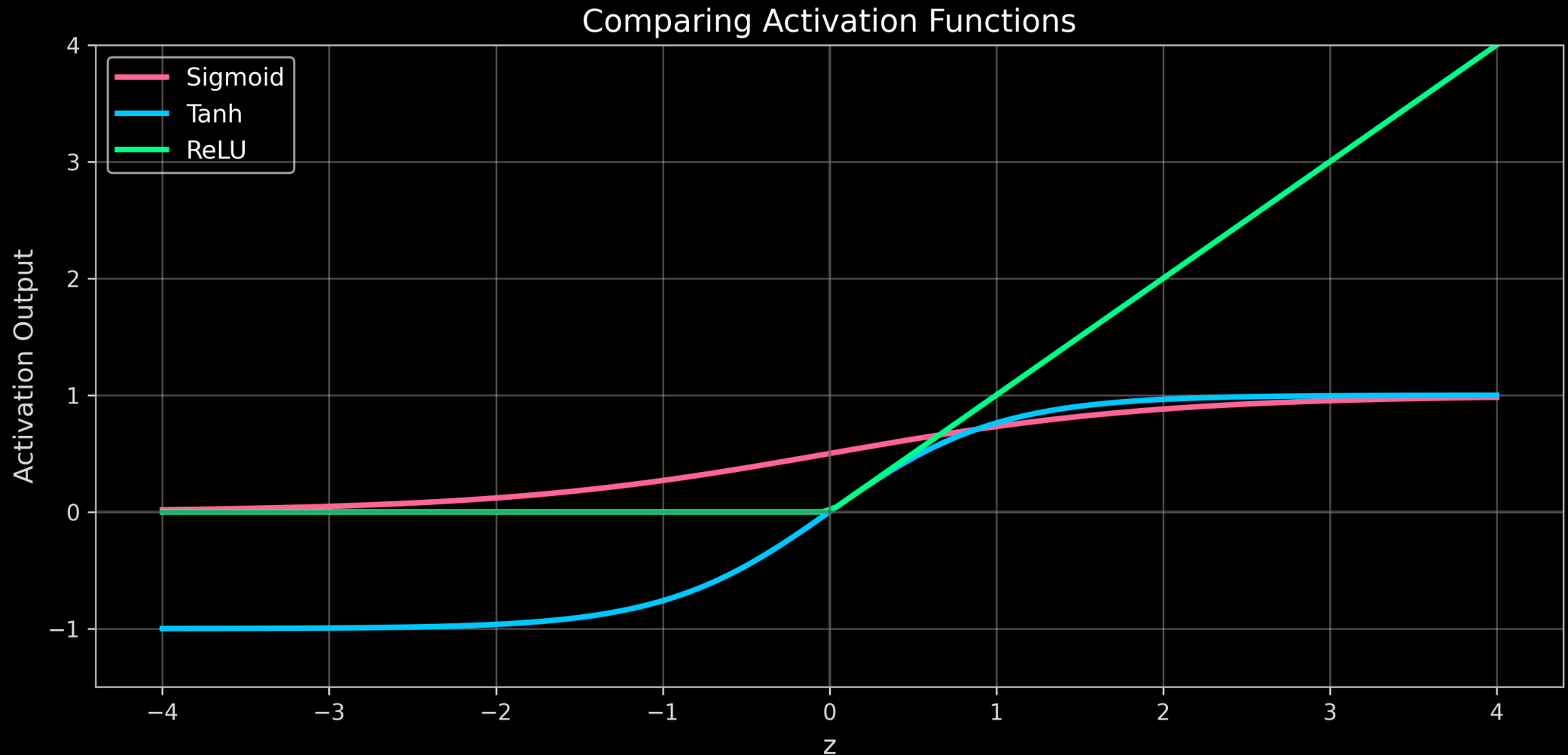
The ReLU Activation Function

$$\text{ReLU}(z) = \max(0, z)$$



Properties: Simple • Fast to compute • Gradient is 0 or 1 (no vanishing!)

Comparing Activation Functions



Choosing Activation Functions

Location	Recommended Activation	Why
Hidden layers	ReLU	Avoids vanishing gradients
Binary output	Sigmoid	Outputs probability (0, 1)
Multi-class output	Softmax	Outputs probabilities that sum to 1
Regression output	None (linear)	Can output any value

What We've Covered

In this video, we've learned:

- Without non-linear activations, deep networks collapse to linear models
- Sigmoid: smooth, outputs $(0,1)$, but has vanishing gradient problem
- Tanh: zero-centered $(-1,1)$, but also has vanishing gradients
- ReLU: simple, fast, gradient is 0 or 1, enables deep networks
- Choose ReLU for hidden layers, sigmoid/softmax for classification outputs